

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 17
 - Continuity and Algebra
 - Connectedness
 - Uniform continuity

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 - Continuity and Algebra
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 - Uniform continuity

Warm-up

Prove that the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = x + y$$

is continuous. (Everything Euclidean)

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$$f(x, y) = xy$$

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Prove that the function $f : \mathbb{R} \times (\mathbb{R} \setminus \{0\}) \rightarrow \mathbb{R}$ defined by

$$f(x, y) = x/y$$

is continuous. (Everything Euclidean)

Algebra and continuity

Theorem

Let (S, d) be a metric space and $f : S \rightarrow \mathbb{R}$ and $g : S \rightarrow \mathbb{R}$ be functions which are continuous at $a \in S$.

- *$f + g$ is continuous at a*
- *fg is continuous at a*
- *f/g is continuous at a as long as $g(a) \neq 0$*

Polynomials and Rational Functions

- constant functions are continuous
- $f(x) = x$ is continuous
- monomials x^n are continuous
- polynomials $a_0 + a_1x + \cdots + a_nx^n$ are continuous
- rational functions $p(x)/q(x)$ are continuous on their domains

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 - **Connectedness**
 - Uniform continuity

Connected spaces

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Examples:

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- the real line with the discrete metric $(\mathbb{R}, d_{disc})$ is connected
- euclidean space $\mathbb{R}^n$ is connected
- an interval $I \subseteq \mathbb{R}$ with the Euclidean metric is connected

Challenge!

Problem

Prove that the set $\mathbb{Q} \subseteq \mathbb{R}$ is disconnected.

Real line is connected

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In particular, U is a nonempty, proper subset of $\mathbb{R}$ which is both open and closed. □

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$$U = (a_1, b_1) \cup (a_2, b_2) \cup \dots$$

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In particular, b_1 is an accumulation point of U but not an element of U .

This contradicts the fact that U is closed. □

Images of connected sets

Theorem

Let (S, d_S) and (T, d_T) be metric spaces and $f : S \rightarrow T$ be a function.

If f is continuous and S is connected, then $f(S)$ is connected.

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Then $f(S) = U \cup V$ for nonempty, disjoint, open sets
 $U, V \subseteq f(S)$.

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Then $f(S) = U \cup V$ for nonempty, disjoint, open sets
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Then $f^{-1}(U)$ and $f^{-1}(V)$ are non-empty, disjoint, open sets and

$$f^{-1}(U) \cup f^{-1}(V) = f^{-1}(U \cup V) = f^{-1}(f(S)) = S.$$

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This is a contradiction. □

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We can make any interval as a continuous image:

$$f : \mathbb{R} \rightarrow \mathbb{R}$$
$$f(x) = \frac{b+a}{2} + \frac{(b-a)x}{1+x^2}$$

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$$f(\mathbb{R}) = (a, b), \quad a < b$$



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We can make any interval as a continuous image:

$$f : \mathbb{R} \rightarrow \mathbb{R}$$

$$f(x) = a + \frac{2(b-a)x^2}{1+x^4}$$

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$$f(\mathbb{R}) = (b, a], \quad a > b$$

Intermediate value theorem

Theorem (Intermediate Value Theorem)

*Suppose $f : [a, b] \rightarrow \mathbb{R}$ is a continuous function.
Then for any value y between $f(a)$ and $f(b)$, there exists
 $c \in [a, b]$ with $f(c) = y$.*

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Without loss of generality, assume $f(a) < f(b)$.

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Then $J = U \cup V$ for

$$U = (-\infty, y) \cap J \quad \text{and} \quad V = (y, \infty) \cap J$$

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These are nonempty, disjoint, and open in the subspace J of $\mathbb{R}$.

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Uniform continuity

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Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$. We say that a function $f : A \rightarrow T$ is **uniformly continuous** on A if for all $\epsilon > 0$ there exists $\delta > 0$ such that

$$d_S(x, y) < \delta \Rightarrow d_T(f(x), f(y)) < \epsilon$$

for all $x, y \in A$.

- f continuous on A means f is continuous at every $a \in A$
- δ can depend on ϵ and a (not on x)
- with uniform continuity, δ can't depend on a ...

Examples

- x is uniformly continuous on $\mathbb{R}$
- x^2 is NOT uniformly continuous on $\mathbb{R}$
- $1/x$ is continuous on $(0, 1)$
- $1/x$ is NOT uniformly continuous on $(0, 1)$

Relationship with compactness

Theorem (Heine)

Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$ be compact. Suppose that $f : A \rightarrow T$ is continuous on A , then f is uniformly continuous on A .